Architecture diagrams for travel

Aircraft Predictive Maintenance



Copyright © 2026 Amazon Web Services, Inc. and/or its affiliates. All rights reserved.

Aircraft Predictive Maintenance: Architecture diagrams for travel

Copyright © 2026 Amazon Web Services, Inc. and/or its affiliates. All rights reserved.

Amazon's trademarks and trade dress may not be used in connection with any product or service that is not Amazon's, in any manner that is likely to cause confusion among customers, or in any manner that disparages or discredits Amazon. All other trademarks not owned by Amazon are the property of their respective owners, who may or may not be affiliated with, connected to, or sponsored by Amazon.

Table of Contents

Aircraft predictive maintenance i
Aircraft predictive maintenance diagram 1
Further reading 2
Diagram history 2

Aircraft Predictive Maintenance

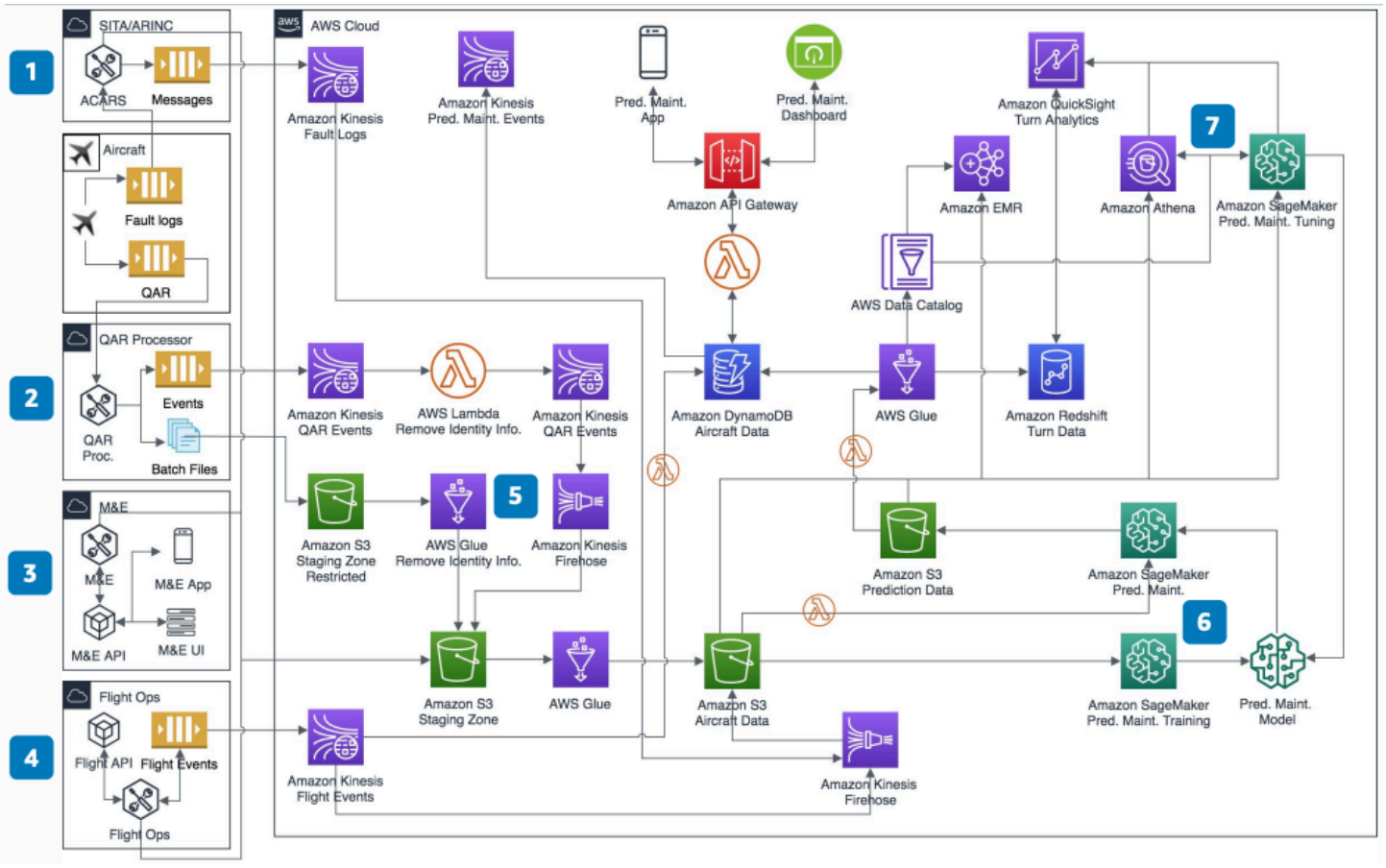
Publication date: **March 10, 2021 (Diagram history)**

Flight delays and cancellations caused by unscheduled maintenance cost airlines USD $120K to USD $300K per aircraft per year. This potentially translates to USD $60M to USD $150M annually for an airline with 500 aircraft.

Implementing predictive maintenance solutions that use aircraft log, sensor, and maintenance data can reduce this cost by up to 25 percent.

This architecture uses Amazon SageMaker AI to correlate aircraft fault data with maintenance history. The trained models predict component failures before they cause delays.

Aircraft predictive maintenance diagram



The following steps describe the architecture:

1. The Aircraft Communications Addressing and Reporting System (ACARS) collects fault logs and pilot reports in real time.
2. The Quick Access Recorder (QAR) collects and uploads data into AWS based on aircraft capability. Frequency varies based on the connectivity package installed.
3. Maintenance and engineering systems provide maintenance logs, part removals, and part repair history periodically.
4. The flight operations system provides scheduled and actual flight information. This correlates delays and cancellations with component failures.
5. You must delete identifying information from QAR data before use.
6. SageMaker AI trains the model to correlate delays and cancellations to fault logs, maintenance logs, part removals, and QAR data. Feature engineering identifies the most significant and predictable chapters and components.
7. Tune models periodically to improve predictions and reduce false positives.

Further reading

For additional information, see the following resources:

- AWS Architecture Icons
- AWS Architecture Center
- AWS Well-Architected

Diagram history

To receive updates about this reference architecture diagram, subscribe to the RSS feed.

Change	Description	Date
Initial publication	Reference architecture diagram first published.	March 10, 2021

RSS subscription requirement

To subscribe to RSS updates, you must have an RSS plugin enabled for the browser you are using.